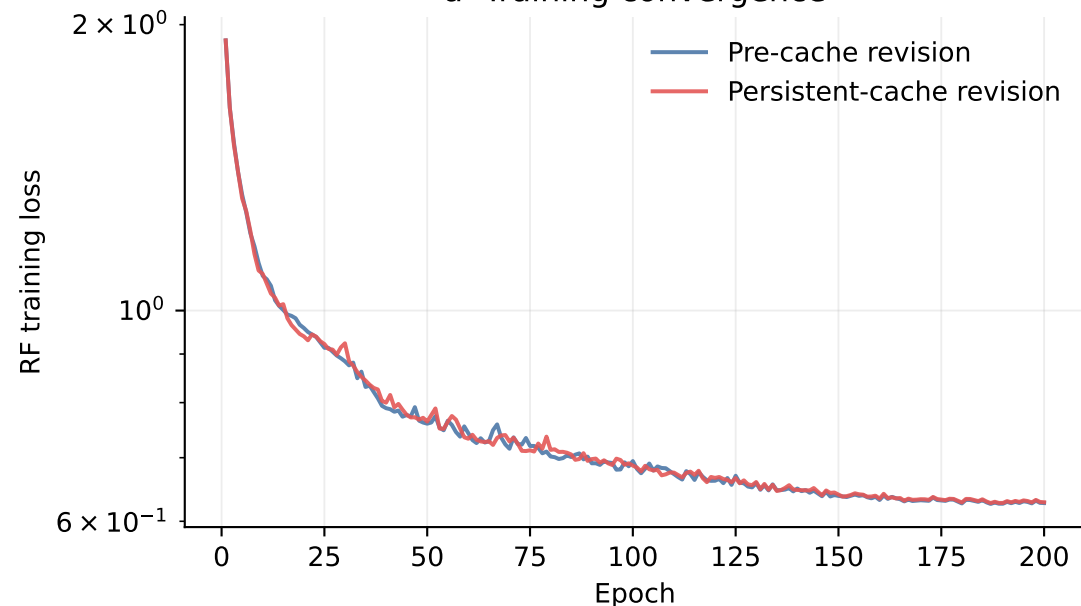
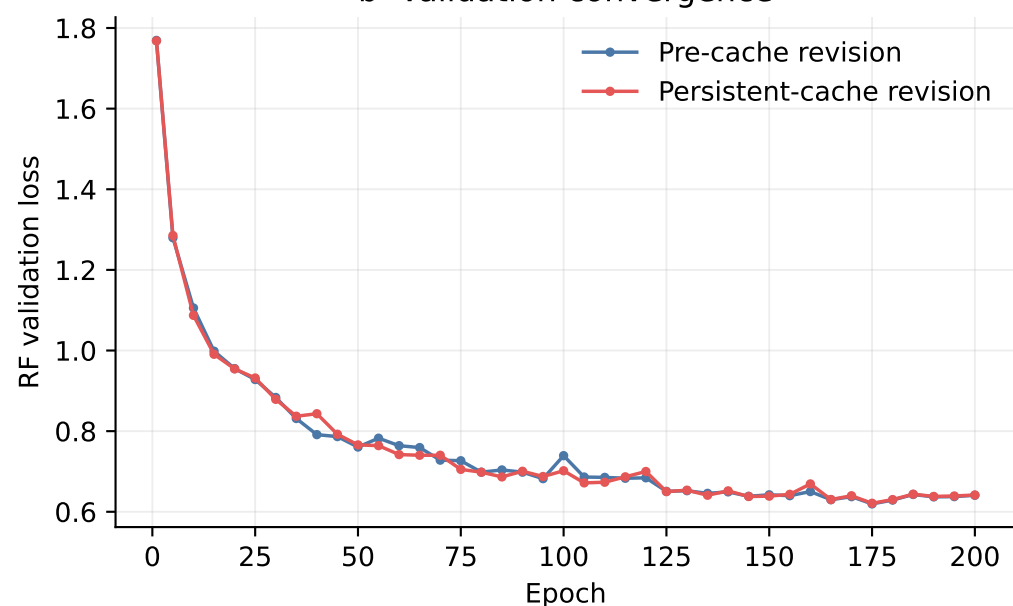


# CQ-LR persistent Top-K: quality neutrality and reconstruction efficiency

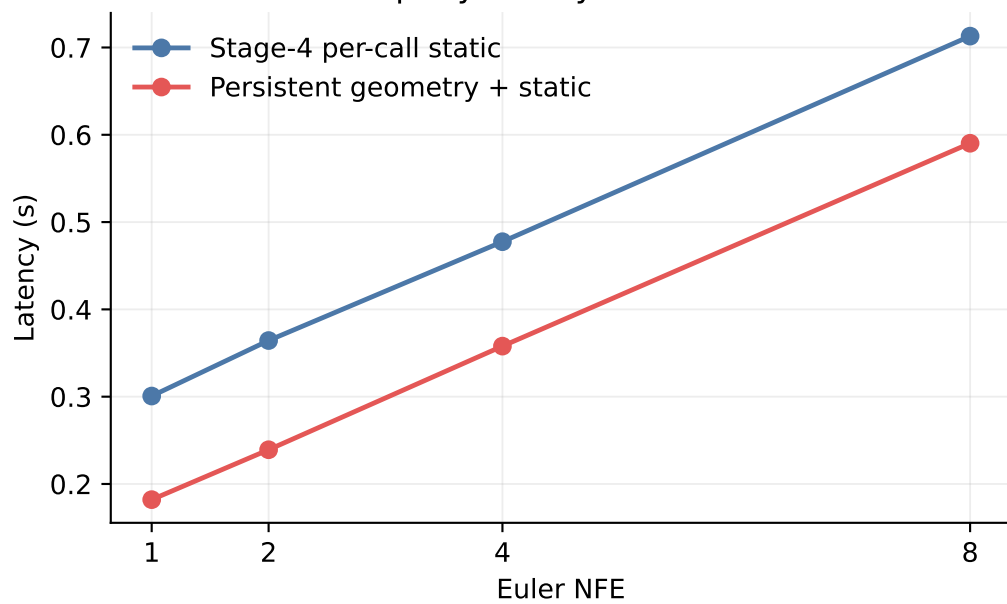
a Training convergence



b Validation convergence



c 1M-query steady reconstruction



d Persistent-cache speedup

